

WPA2 *Wi-Fi Protected Access 2 (WPA2)* uses the *Advanced Encryption Standard (AES)* to provide a secure wireless environment. All current WAPs and wireless clients support WPA2. You may encounter older routers with a “backward compatible” mode for first-generation WPA—but at this point I recommend upgrading or replacing WPA-only devices if at all possible.



EXAM TIP WPA2 is also compatible with the *Temporal Key Integrity Protocol (TKIP)*, an older encryption method that was introduced with the original WPA. You may see it as an option (both on the exam and in WPA2 devices), but—as an option for backward compatibility with even older devices—enabling it will make a modern network less secure (and potentially slow it down).

WPA3 The successor to WPA2 was announced in early 2018. *Wi-Fi Protected Access 3 (WPA3)* addresses some security and usability issues, including encryption to protect the data of users on open (public) networks.



NOTE WPA2 was the best option we had for well over a decade. The transition to WPA3 is under way, but WAPs tend to have a pretty long life—expect WPA2 to be the best option on many older devices. Implement WPA3 networks when practical, and eventually disable WPA2 once all of the devices you need to support are WPA3-compatible.

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Wireless Networking Standards and Regulations

Most wireless networks use *radio frequency (RF)* technologies, in particular the 802.11 (Wi-Fi) standards. (Other standards such as Bluetooth hold a much smaller place in today's market.) These technologies use parts of the RF signal spectrum that were already in use for other purposes, so they are tightly regulated to minimize their impact on those users. To help you gain a better understanding of wireless network technologies, this section provides a brief look at the standards they use and some aspects of how regulations come into play.



NOTE Radio frequency is the part of the electromagnetic spectrum used for radio communication.

IEEE 802.11-Based Wireless Networking

The IEEE 802.11 wireless Ethernet standard, more commonly known as *Wi-Fi*, defines methods devices may use to communicate via *spread-spectrum* radio waves. Spread-spectrum broadcasts data in small, discrete chunks over the frequencies available within a certain frequency range.